

LOGIC (Proposition)

Sr. No.	Connective	Symbol	Name of corresponding Compound sentence
1.	AND	$\wedge$	Conjunction
2.	OR	$\vee$	Disjunction
3.	NOT	$\sim$	Negation
4.	IF...THEN	$\rightarrow$	Conditional
5.	IFF ↓ If & only if	$\leftrightarrow$	Biconditional

IF...THEN

P	Q	$P \rightarrow Q$
T	T	T
T	F	F
F	T	T
F	F	T

only F when 2nd value is F
otherwise T.

IFF (Like XNOR)

P	Q	$P \leftrightarrow Q$
T	T	T
T	F	F
F	T	F
F	F	T

Both values same $\rightarrow$ T
Otherwise $\rightarrow$ F

Others are completely same as boolean algebra,
AND, OR, NOT.

Converse, Inverse, Contra positive

d. If $[p \rightarrow q]$ is a conditional statement.

Converse: $q \rightarrow p$

Inverse: $\sim p \rightarrow \sim q$

Contra positive: $\sim q \rightarrow \sim p$

Example 3.4.4 Express following statement in propositional form :

- i) There are many clouds in the sky but it did not rain.
- ii) I will get first class if and only if I study well and score above 80 in mathematics.
- iii) Computers are cheap but softwares are costly.
- iv) It is very hot and humid or Ramesh is having heart problem.
- v) In small restaurants the food is good and service is poor.

(i) p : There are many cloud in the sky
 q : It rain

Symbolic form: $p \wedge \sim q$
 $\downarrow$
but/and

(ii) p : I will get first class
 q : I study well
 r : I score above 80 in math.

$p \leftrightarrow (q \wedge r)$

(iii) $q \wedge r$

(iv) $(p \wedge q) \vee r$

(v) $p \wedge q$

Example 3.4.6 Express the contrapositive, converse, inverse and negation forms of the conditional statement given below.

'If x is rational, then x is real'.

$(p \rightarrow q)$: Symbolic form

$(q \leftarrow p)$: If x is real then x is rational. (converse)

$(\sim p \rightarrow \sim q)$: If x is not rational then x is not real. (Inverse)

$(\sim q \rightarrow \sim p)$: If x is not real then x is not rational.
 (Contrapositive)

Example 3.4.5 Express the contrapositive, converse and inverse forms of the following statement if $3 < b$ and $1 + 1 = 2$, then $\sin \frac{\pi}{3} = \frac{1}{2}$.

if $3 < b$ and $1 + 1 = 2$, then $\sin \frac{\pi}{3} = \frac{1}{2}$

Let, p : $3 < b$

q : $1 + 1 = 2$

r : $\sin \frac{\pi}{3} = \frac{1}{2}$

Symbolic form: $(p \wedge q) \rightarrow r$

Converse: $r \leftarrow (p \wedge q)$ If $\sin \frac{\pi}{3} = \frac{1}{2}$ then $3 < b$ and $1 + 1 = 2$.

Inverse: $\sim(p \wedge q) \rightarrow \sim r$

$= (\sim p \vee \sim q) \rightarrow \sim r$

if $3 \geq b$ or $1 + 1 \neq 2$ then $\sin \frac{\pi}{3} \neq \frac{1}{2}$.

Contrapositive: $\sim r \leftarrow \sim(p \wedge q)$
 $= \sim r \leftarrow (\sim p \vee \sim q)$

If $\sin \frac{\pi}{3} \neq \frac{1}{2}$ then $3 \geq b$ and $1 + 1 \neq 2$.

Determine whether each of the following statement formula is a tautology, contradiction or contingency. $(p \wedge q) \wedge \sim(p \vee q)$

p	q	$p \wedge q$	$p \vee q$	$\sim(p \vee q)$	$(p \wedge q) \wedge \sim(p \vee q)$
T	T	T	T	F	F
T	F	F	T	F	F
F	T	F	T	F	F
F	F	F	F	T	F

Tautology
 all values are T
Contradiction
 all values F
Contingency
 Mix of T & F
 Statement Pattern
 → Contradiction

$$(p \rightarrow q) \leftrightarrow (q \vee \sim p)$$

p	q	$p \rightarrow q$	$\sim p$	$q \vee \sim p$	$(p \rightarrow q) \leftrightarrow (q \vee \sim p)$
T	T	T	F	T	T
T	F	F	F	F	F
F	T	T	T	T	T
F	F	T	T	T	T

Tautology

$$(p \rightarrow (q \rightarrow r)) \rightarrow ((p \rightarrow q) \rightarrow (p \rightarrow r))$$

p	q	r	$p \rightarrow q$	$q \rightarrow r$	$p \rightarrow r$	$p \rightarrow (q \rightarrow r)$	$(p \rightarrow q) \rightarrow (p \rightarrow r)$	$(p \rightarrow (q \rightarrow r)) \rightarrow ((p \rightarrow q) \rightarrow (p \rightarrow r))$
T	T	T	T	T	T	T	T	T
T	T	F	T	F	F	F	F	T
T	F	T	F	T	T	T	T	T
T	F	F	F	T	F	T	T	T
F	T	T	T	T	T	T	T	T
F	F	T	T	T	T	T	T	T
F	T	F	T	F	F	T	T	T
F	F	F	T	T	T	T	T	T

Tautology

Prove by Constructing truth Table $p \rightarrow (q \vee r) \equiv (p \rightarrow q) \vee (p \rightarrow r)$.

p	q	r	$q \vee r$	$p \rightarrow (q \vee r)$	$p \rightarrow q$	$p \rightarrow r$	$(p \rightarrow q) \vee (p \rightarrow r)$	$p \rightarrow (q \vee r) \equiv (p \rightarrow q) \vee (p \rightarrow r)$
T	T	T	T	T	T	T	T	✓
T	T	F	T	T	T	F	T	✓
T	F	T	T	T	F	T	T	✓
T	F	F	F	F	F	F	F	✓
F	T	T	T	T	T	T	T	✓
F	T	F	T	T	T	F	T	✓
F	F	T	T	T	T	T	T	✓
F	F	F	F	T	T	T	T	✓

Proved.

Idempotence of $\vee$	$p \equiv p \vee p$
Idempotence of $\wedge$	$p \equiv p \wedge p$
Commutativity of $\vee$	$p \vee q \equiv q \vee p$
Commutativity of $\wedge$	$p \wedge q \equiv q \wedge p$
Associativity of $\vee$	$p \vee (q \vee r) \equiv (p \vee q) \vee r$

Associativity of $\wedge$	$p \wedge (q \wedge r) \equiv (p \wedge q) \wedge r$
Distributivity of $\wedge$ over $\vee$	$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$
Distributivity of $\vee$ over $\wedge$	$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$
Double negation	$p \equiv \sim(\sim p)$
De Morgan's laws	$\sim(p \vee q) \equiv \sim p \wedge \sim q$
De Morgan's laws	$\sim(p \wedge q) \equiv \sim p \vee \sim q$
Tautology	$p \vee \sim p \equiv \text{Tautology}$
Contradiction	$p \wedge \sim p \equiv \text{Contradiction}$
Absorption laws	$p \vee (p \wedge q) \equiv p$
Absorption laws	$p \wedge (p \vee q) \equiv p$

Imp.

$$p \vee \sim p \equiv T$$

$$p \wedge \sim p \equiv F$$

$$p \rightarrow q \equiv \sim p \vee q$$

$$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$$

identity law

$$p \vee T \equiv T$$

$$p \vee F \equiv p$$

$$p \wedge T \equiv p$$

$$p \wedge F \equiv F$$

Disjunctive Normal form (dnf)

→ A statement form which consist of Disjunction between conjunction is called DNF.

$$\text{eg. } (P \wedge Q) \vee Q \vee (\sim Q \wedge P)$$

Conjunction
(And)

Disjunction
(OR)

$$(\wedge) \vee (\wedge) \vee (\wedge) \vee \dots$$

conjunctive Normal Form (cnf)

→ A statement form which consist of conjunction between disjunction is called CNF.

$$\text{eg. } (P \vee Q) \wedge (\sim P \vee Q) \wedge (\sim Q)$$

disjunction

conjunction.

Q. Obtain DNF: $(P \rightarrow Q) \wedge (Q \rightarrow P)$

$$(P \rightarrow Q) \wedge (Q \rightarrow P)$$

$$\equiv (\sim P \vee Q) \wedge (\sim Q \vee P) \quad [\because (P \rightarrow Q) \equiv (\sim P \vee Q)]$$

$$\equiv [(\sim P \vee Q) \wedge \sim Q] \vee [(\sim P \vee Q) \wedge P] \quad [\text{distributive law}]$$

$$\equiv [(\sim P \wedge \sim Q) \vee (Q \wedge \sim Q)] \vee [(\sim P \wedge P) \vee (Q \wedge P)]$$

$$\equiv [(\sim P \wedge \sim Q) \vee F] \vee [F \vee (P \wedge Q)] \quad [\text{Commutative law}]$$

$$\equiv [(\sim P \wedge \sim Q)] \vee [(P \wedge Q) \vee F] \quad [\text{Commutative law, } P \vee F = P]$$

$$\equiv \boxed{(\sim P \wedge \sim Q) \vee (P \wedge Q)} \quad [P \vee F = P]$$

disjunction achieved. → Ans.

Q. Find DNF: $(P \wedge (P \rightarrow Q)) \rightarrow Q$

$$(P \wedge (P \rightarrow Q)) \rightarrow Q$$

$$\equiv \sim(P \wedge (P \rightarrow Q)) \vee Q \quad [\because P \rightarrow Q = \sim P \vee Q]$$

$$\equiv \sim(P \wedge (\sim P \vee Q)) \vee Q$$

$$\equiv (\sim P \vee \sim(\sim P \vee Q)) \vee Q$$

$$\equiv (\sim P \vee (P \wedge \sim Q)) \vee Q$$

$$\equiv \underline{\sim P} \vee \underline{(P \wedge \sim Q)} \vee \underline{Q} \rightarrow \underline{\text{Ans.}}$$

Q. Obtain CNF: $p \wedge (p \rightarrow q)$

$$\begin{aligned}
 (i) & p \wedge (p \rightarrow q) \\
 & \equiv p \wedge (\sim p \vee q) \\
 & \equiv (p \wedge \sim p) \vee (p \wedge q) \\
 & = F \vee (p \wedge q) \\
 & = (p \wedge q) \vee F \text{ [Commutative]} \\
 & = p \wedge q \text{ [}\because p \vee F = p\text{]}
 \end{aligned}$$

Ans.

$$\begin{aligned}
 (ii) & q \vee (p \wedge \sim q) \vee (\sim p \wedge \sim q) \\
 & \equiv (q \vee p) \wedge (q \vee \sim q) \vee \sim(p \vee q) \\
 & \equiv (p \vee q) \wedge T \vee \sim(p \vee q) \\
 & \equiv (p \vee q) \vee \sim(p \vee q) \\
 & \equiv T
 \end{aligned}$$

Ans.

Q. Obtain DNF: $((p \rightarrow q) \wedge (q \rightarrow p)) \vee p$

$$\begin{aligned}
 & ((p \rightarrow q) \wedge (q \rightarrow p)) \vee p \\
 & \equiv (\sim p \vee q) \wedge (\sim q \vee p) \vee p \\
 & \equiv ((\sim p \vee q) \vee p) \wedge ((\sim q \vee p) \vee p) \\
 & \equiv ((\sim p \vee p) \vee q) \wedge ((\sim q \vee p) \vee p) \\
 & \equiv [T \vee q] \wedge (\sim p \vee p) \\
 & \equiv T \wedge (\sim p \vee p) \\
 & \equiv (\sim p \vee p) \wedge T \\
 & \equiv \sim p \vee p
 \end{aligned}$$

Ans.

Valid Arguments : $p_1, p_2, p_3, \dots, p_n$ (Premises)

$p_1 \wedge p_2 \wedge \dots \wedge p_n \rightarrow c$ is a Tautology

$$\begin{array}{l}
 S_1: p_1 \\
 S_2: p_2 \\
 \hline
 S_3: c
 \end{array}$$

$$p_1 \wedge p_2 \rightarrow c$$

Modus Ponens

$p \rightarrow q$ as true
 p as true
 $\therefore q$??

$$\begin{array}{l}
 S_1: p \rightarrow q \\
 S_2: p \\
 \hline
 S_3: q
 \end{array}$$

$(p \rightarrow q) \wedge p \rightarrow q$ is a Tautology

P	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Modus Tollens

$p \rightarrow q$ as true
 q as false
 p is ___?

$$\begin{array}{l}
 S_1: p \rightarrow q \\
 S_2: \sim q \\
 \hline
 S_3: \sim p
 \end{array}$$

$(p \rightarrow q) \wedge \sim q \rightarrow \sim p$
 is a Tautology

Hypothetical Syllogism

$S_1: p \rightarrow q$ as true

$S_2: q \rightarrow r$ as true

$S_3: p \rightarrow r$

$(p \rightarrow q) \wedge (q \rightarrow r) \rightarrow (p \rightarrow r)$
is a Tautology

	p	q	$p \vee q$
X	T	T	T
X	T	F	T
✓	F	T	T
	F	F	F

Disjunctive Syllogism

$p \vee q$ as true

p as False

q is T?

$S_1: p \vee q$

$S_2: \sim p$

$S_3: q$

$(p \vee q) \wedge \sim p \rightarrow q$ is a
tautology

Test the validity of the argument. If a person is poor, he is unhappy. If a person is unhappy, he dies young. Therefore poor person dies young.

let, p: person is poor.

q: person is unhappy.

r: person dies young.

Symbolic Form

$S_1: p \rightarrow q$

$S_2: q \rightarrow r$

$S_3: p \rightarrow r$

Above argument is valid according to Hypothetical Syllogism.

Determine whether the argument is valid or not. If I try hard and I have talent then I will become a musician. If I become a musician, then I will be happy. Therefore if I will not be happy then I did not try hard or I do not have talent.

let, p: I try hard

q: I have talent

r: I will become a musician.

s: I will be happy.

Symbolic Form

$S_1: (p \wedge q) \rightarrow r$

$S_2: r \rightarrow s$

$S_3: \sim s \rightarrow (\sim p \vee \sim q)$

T
T
F

Suppose the given Argument is invalid.

$\therefore S_1$ as T, S_2 as T, S_3 as F

$T \rightarrow F \equiv F$

$F \vee F \equiv F$

$\sim p \equiv F$

$p \equiv T$

$\sim q \equiv F$

$q \equiv T$

$\sim s \rightarrow (\sim p \vee \sim q)$ (F)
 $T \rightarrow F$ (F)

$s \equiv F, p \equiv T, q \equiv T$

$$S_2 =$$

$r \rightarrow s$	(T)
$(F) \rightarrow F$	(T)

$r \equiv F$

$$S_1: P \wedge q \rightarrow r$$

$$T \wedge T \rightarrow F$$

$$T \rightarrow F \equiv (F)$$

But given True.

✓ here Argument is valid Proved

Quantifiers.

Universal
Quantifiers

$\forall \Rightarrow$ for all

Existential
Quantifiers

$\exists \Rightarrow$ There exists

Predicate:

eg. x is a prime no.

Negation of Quantified Statement

$\forall x P(x)$	$\exists x [\sim P(x)]$
$\exists x [\sim P(x)]$	$\forall x P(x)$
$\forall x [\sim P(x)]$	$\exists x P(x)$
$\exists x P(x)$	$\forall x [\sim P(x)]$

Let $p(x)$: x is even $\rightarrow$ $\sim$ odd.

$Q(x)$: x is a prime number

$R(x, y)$: $x + y$ is even

a) Using the information given above write the following sentences in symbolic form.

- Every integer is an odd integer
- Every integer is even or prime
- The sum of any two integers is an odd integer.

- $\forall x [\sim P(x)]$
- $\forall x [P(x) \vee Q(x)]$
- $\forall x \forall y [\sim R(x, y)]$

Negate each of the following statements

- i) $\forall x, |x| = x$ ii) $\exists x, x^2 = x$

(i) $\forall x, |x| = x$

$\Rightarrow \exists x, |x| \neq x$

(ii) $\exists x, x^2 = x$

$\Rightarrow \forall x, x^2 \neq x$

Negate the following

- i) If there is a riot, then someone is killed.
ii) It is day light and all the people are arisen.

It is not the case that Copy the sentence

Fixed in all cases.

- Q. $Q(x)$: x is a prime no.
 $P(x)$: x is even
 $R(x, y)$: $(x+y)$ is even.

Convert the symbols to sentences:

① $\forall x (\sim Q(x))$

② $\exists y (\sim P(y))$

③ $\sim [\exists x (P(x) \wedge Q(x))]$

Ans.

① All the integers are not prime numbers.

② Atleast one integer is not even.

③ It is not case that there exists an integer which is even and prime.